Basic Google Tag Manager Setup Plan

Digital Marketing Track | Sadhanala Nischay Prasad | Tools: Google Tag Manager, Google Analytics 4

1. Objective

Plan the tags, triggers and variables needed to track important website interactions (button clicks, form submissions and file downloads) and send them to GA4, and understand the fundamentals of tag management. The plan is built for a demo landing page of a fictional Hyderabad tea brand, *Charminar Chai Co.*, included in the GitHub repository.

2. Fundamentals

Concept	Meaning	Example
Tag	Code that sends data to a tool such as GA4	GA4 - Event - file_download
Trigger	Rule that decides when a tag fires	Click - File Downloads
Variable	Value GTM reads when the event happens	{{Click URL}}, {{CJS - File Extension}}
Data layer	JavaScript array the site uses to pass information to GTM	dataLayer.push({event: 'form_submit_success'})

3. Measurement goals

Following the rule "define what you want to measure before creating tags", each tag answers a business question.

Business question	What is measured	GA4 event
Are visitors interested enough to act?	CTA button clicks	cta_click
Is the catalogue useful?	Catalogue PDF downloads	file_download
How many real leads do we get?	Successful enquiry form submissions	generate_lead (key event)
Do people contact us directly?	Phone, WhatsApp and email clicks	contact_click (key event)
Where do visitors leave to?	Clicks to other websites	outbound_click
Do people read the full page?	Scroll depth 50% / 90%	scroll_depth

4. Tag plan

#	Tag name	Type	Parameters	Trigger
T1	GA4 - Google Tag - Config	Google Tag	Measurement ID via {{Const - GA4 Measurement ID}}; sends page_view	Initialization - All Pages
T2	GA4 - Event - cta_click	GA4 Event	cta_text={{Click Text}}, cta_location={{AEV - CTA Location}}, page_path={{Page Path}}	Click - CTA Buttons
T3	GA4 - Event - file_download	GA4 Event	file_name, file_extension, link_url, link_text	Click - File Downloads
T4	GA4 - Event - generate_lead	GA4 Event	form_id, form_name, enquiry_type (data layer)	Custom Event - form_submit_success
T5	GA4 - Event - contact_click	GA4 Event	contact_method={{RT - Contact Method}}, link_url	Click - Contact Links
T6	GA4 - Event - outbound_click	GA4 Event	link_url, link_domain={{URL - Click Hostname}}, link_text	Click - Outbound Links
T7	GA4 - Event - scroll_depth	GA4 Event	percent_scrolled={{Scroll Depth Threshold}}	Scroll - 50 / 90

5. Trigger plan

#	Trigger name	Type	Conditions
R1	Initialization - All Pages	Initialization	All pages
R2	Click - CTA Buttons	Click - All Elements	{{Click Classes}} contains js-cta
R3	Click - File Downloads	Click - Just Links	{{Click URL}} matches RegEx (ignore case) \.(pdf|docx?|xlsx?|pptx?|zip|csv)($|\?)
R4	Custom Event - form_submit_success	Custom Event	Event name equals form_submit_success (pushed only after validation passes)
R5	Click - Contact Links	Click - Just Links	{{Click URL}} matches RegEx ^(tel:|mailto:|https://wa\.me/)
R6	Click - Outbound Links	Click - Just Links	{{Click URL}} starts with http AND {{URL - Click Hostname}} does not contain {{Page Hostname}} AND {{Click URL}} does not match wa\.me
R7	Scroll - 50 / 90	Scroll Depth	Vertical, percentages 50,90, all pages

6. Variable plan

Built-in variables enabled: Page URL, Page Hostname, Page Path, Referrer, Event, Click Element, Click Classes, Click ID, Click Target, Click URL, Click Text, Form Element, Form ID, Form Classes, Scroll Depth Threshold, Scroll Depth Units, Scroll Direction.

#	Variable	Type	Configuration
V1	Const - GA4 Measurement ID	Constant	G-XXXXXXXXXX
V2	DLV - form_id	Data Layer Variable	form_id (version 2)
V3	DLV - form_name	Data Layer Variable	form_name (version 2)
V4	DLV - enquiry_type	Data Layer Variable	enquiry_type (version 2)
V5	AEV - CTA Location	Auto-Event Variable	Element Attribute: data-cta-location
V6	CJS - File Extension	Custom JavaScript	Extension from {{Click URL}}, e.g. pdf
V7	CJS - File Name	Custom JavaScript	Last path segment of {{Click URL}}
V8	URL - Click Hostname	URL	Component: Host Name, source {{Click URL}}
V9	RT - Contact Method	Regex Table	^tel: = phone, ^mailto: = email, wa\.me = whatsapp, default other

7. Event tracking map

#	User action	Element	GA4 event	Parameters	Key
E1	Opens any page	-	page_view	page_location, page_title	No
E2	Clicks "Order a box"	#cta-order.js-cta	cta_click	cta_text, cta_location, page_path	No
E3	Clicks "Wholesale enquiry"	#cta-wholesale.js-cta	cta_click	cta_text, cta_location, page_path	No
E4	Downloads catalogue PDF	a[href$=".pdf"]	file_download	file_name, file_extension, link_url, link_text	No
E5	Submits enquiry form	form#enquiry-form	generate_lead	form_id, form_name, enquiry_type	Yes
E6	Clicks phone number	a[href^="tel:"]	contact_click	contact_method=phone, link_url	Yes
E7	Clicks WhatsApp	a[href^="https://wa.me/"]	contact_click	contact_method=whatsapp, link_url	Yes
E8	Clicks email	a[href^="mailto:"]	contact_click	contact_method=email, link_url	Yes
E9	Clicks Instagram / Swiggy	external link	outbound_click	link_url, link_domain, link_text	No
E10	Scrolls 50% / 90%	-	scroll_depth	percent_scrolled	No

8. Key decisions

• **Data layer event for leads.** The built-in Form Submission trigger can fire on failed submits; the site pushes form_submit_success only after validation passes.

• **No personal data in GA4.** Names and emails are never sent; only form_id, form_name and enquiry_type.

• **No double counting.** GA4 Enhanced Measurement toggles for outbound clicks, file downloads, scrolls and form interactions are turned off because GTM tracks them.

• **GA4 recommended names** (generate_lead, file_download) are used where they exist.

• **Custom dimensions** (cta_text, cta_location, form_name, enquiry_type, contact_method, link_domain, file_name, percent_scrolled) are registered in GA4 so they appear in reports.

9. Testing and publishing

• GTM Preview (Tag Assistant): perform every action in the map and confirm the correct tag fired and variables have values.

• Submit the form with an empty email: generate_lead must not fire. Submit correctly: it fires once.

• GA4 DebugView: confirm events and parameters arrive.

• Publish version "v1.0 - GA4 base + key interactions", check Realtime, then mark generate_lead and contact_click as key events.

10. Learnings

Tags, triggers and variables are kept separate so they can be reused: one Measurement ID variable feeds every tag, and one trigger can drive several tags. Planning the event map first made the GTM build straightforward and kept event names consistent. The data layer is the bridge for anything GTM cannot see on its own, such as a successful form submission.